



Department of Computer Science and Engineering (CSE)

Trimester : Fall 2025 (253)

Course Code : EEE 2124

Course Name : Electronics Laboratory

Experiment No:01

Experiment Name:Study of Diode I-V Characteristics with Multisim

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Submitted By

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Short Theory

A semiconductor diode is a PN-junction device that allows current to flow only in one direction.

In forward bias, the diode voltage increases until it reaches a threshold voltage (≈ 0.7 V for silicon), after which the current increases rapidly.

In reverse bias, only a very small leakage current flows until the breakdown region.

The diode I–V curve is non-linear, showing exponential current increase in forward bias and very low current in reverse bias.

Objectives

1. To study the current–voltage (I–V) characteristics of a semiconductor diode.
2. To observe the diode behavior in forward and reverse bias conditions.
3. To determine the forward threshold (turn-on) voltage of a silicon diode.
4. To calculate the diode current using the series resistor and Ohm's Law
5. To plot the I–V curve using experimentally measured data.

Equipment Used

DC Power Supply

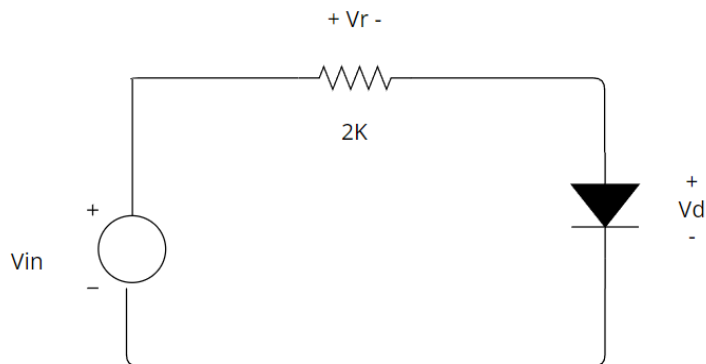
Resistor

Silicon Diode

Multimeter

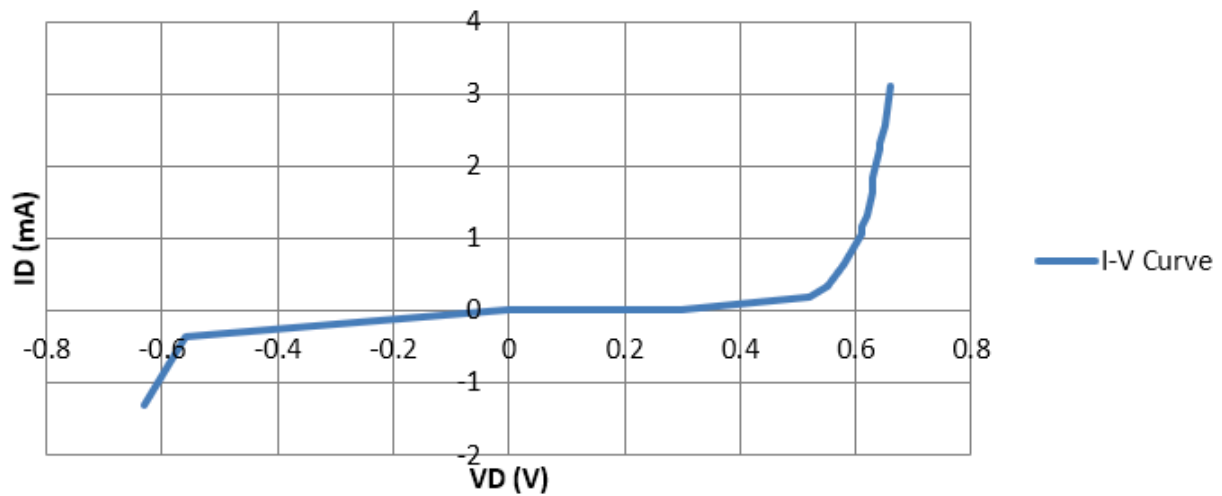
Connecting Wires

Circuit Diagram



Graph

Diode I-V Characteristics



Discussion

The experiment successfully demonstrates the non-linear I-V characteristics of a silicon diode.

Key findings:

- In reverse bias, current is extremely small ($\approx$ micro amps), matching theory.

- In forward bias, current remains near zero until the diode voltage reaches around 0.6–0.7 V.
- After crossing threshold, current increases rapidly — indicating exponential behavior.
- The measured resistor value 0.978k Ω 0.978 k Ω 0.978k Ω was used for accurate current calculation.
- The measured data closely matches the theoretical diode curve from lecture notes.

This verifies the typical behavior of a silicon diode as a one-way current device with a specific turn-on voltage.